



**Name: Mutangana Joseph ID: 29061**

## **Linux Administrator Course**

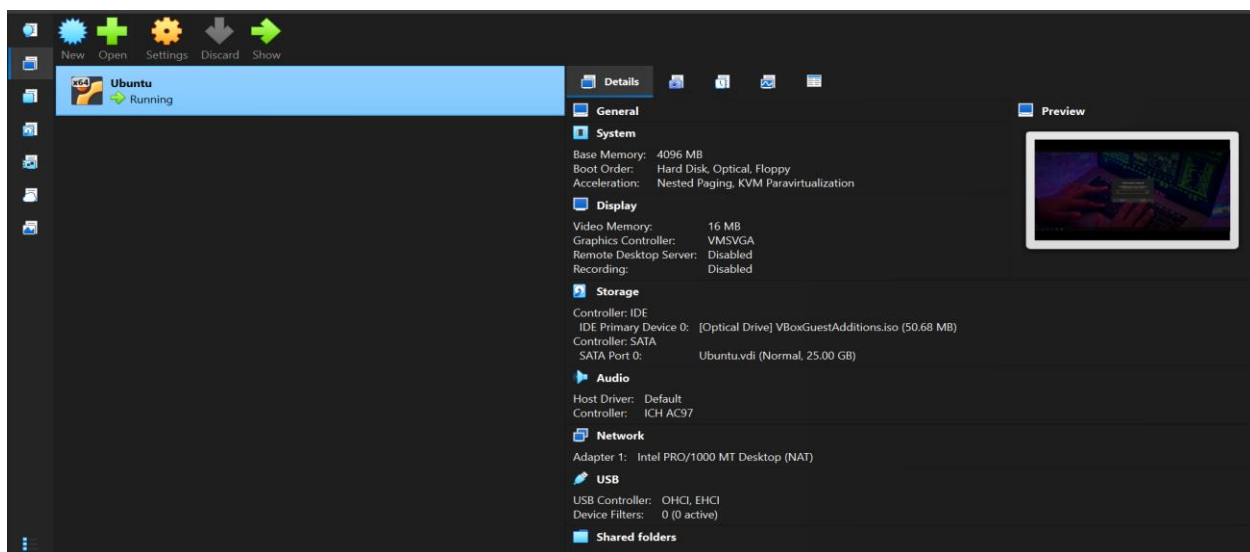
### **Assignment#1 Report**

#### **Task 1: Environment Setup**

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Ubuntu OS was installed using virtual box as virtual machine, it has 4GB of RAM, 2CPU cores and 25GB of disk storage

**Figure 1: VM Configuration**



**Figure 2: OS Version / Release**

Below is the OS version and release of Ubuntu OS I installed as virtual machine

```
mutangana@linux:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 22.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="22.04"
VERSION="22.04.5 LTS (Jammy Jellyfish)"
VERSION_CODENAME=jammy
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=jammy
```



## Task 2: Create Dedicated Student Account

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```
sudo useradd --badname -m -g mutangana -s /bin/bash 29061  
  
sudo passwd 29061  
  
sudo usermod -aG sudo 29061
```

*The command above used to force linux accept bad username like 29061*

**Figure 3:** whoami output

```
29061@linux:~$ whoami  
29061  
29061@linux:~$
```

## Task 3: Terminal Customization

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Terminal was customized to show the date and time in terminal and when command executed the terminal show current time and date beside username and hostname displayed on terminal.

**Figure 4:** Customized terminal output

```
(29061-@linux [2026-09-18 09:45:46])-[~] > whoami  
29061  
(29061-@linux [2026-09-18 09:46:07])-[~] > █
```

## Task 4: Command practice screen-recorded videos demonstration

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The screen-recorded videos were uploaded on google drive, use the link below to access the videos.

[https://drive.google.com/drive/folders/1sP0LSWD5WmOLf7u7xqHvQFOjcoCa5bu?usp=drive\\_link](https://drive.google.com/drive/folders/1sP0LSWD5WmOLf7u7xqHvQFOjcoCa5bu?usp=drive_link)

## Task 5: Shell script

---

The script I created is called semester directory setup (SDS). Why this SDS, because I have been creating same folders every semester manual, example, if I'm semester 5 I create sem-5 inside that directory I create sub directories of courses I'm going to be learning in certain semester, and inside of them, each course, has folders like, notes, assignments, solutions, revisions, something like that.

As you can see, there was repeatedly subdirectories creation in each course directory. With this SDS, I



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can run it and prompt me to enter the name of the semester I'm going to create directory for, and it ask me how many courses am I taking in that semester, then it loops through that number asking me the name of course I'm having and after it create default directories such notes, assignment in each course directory.



**Figure 5:** SDS running in terminal

```
(29061-🌐linux [2026-09-19 01:30:52])-[~/runscript] > sudo bash ../scripts/sds.sh

=====
      WELCOME TO SEMESTER DIRECTORY SETUP (SDS)
=====

Enter your name: Mutangana Joseph
Enter your semester, e.g Fall2026: Sem-6
Success!

1. Tables
2. Payments

Can SDS create above directories in Sem-6? Y/N: y
Success!

How many courses for this semester?: 2

Enter a name for course 1: Linux

1. Notes
2. Assignments
3. Solutions
4. Revisions

Can SDS create above directories in Linux? Y/N: y
Success!

Enter a name for course 2: OOP

1. Notes
2. Assignments
3. Solutions
4. Revisions

Can SDS create above directories in OOP? Y/N: y
Success!
```



Enter a name for course 2: OOP

1. Notes
2. Assignments
3. Solutions
4. Revisions

Can SDS create above directories in OOP? Y/N: y  
Success!

```
*****
Your Sem-6 semester directory setup processed successfully
*****
```

Student Name: Mutangana Joseph  
Semester name: Sem-6  
Number of courses: 2  
Disk usage: 56K .  
Directory path: /home/29061/runscript/Sem-6  
Created on: Sat 19 Sep 2026 01:32:04 SAST

```
(29061-@linux [2026-09-19 01:33:35])-[~/runscript] > sudo ls
Sem-6
(29061-@linux [2026-09-19 01:33:42])-[~/runscript] > cd Sem-6
(29061-@linux [2026-09-19 01:34:11])-[~/runscript/Sem-6] > ls
info Linux OOP Payments Tables
(29061-@linux [2026-09-19 01:34:14])-[~/runscript/Sem-6] > cat info
```

```
*****
Your Sem-6 semester directory setup processed successfully
*****
```

Student Name: Mutangana Joseph  
Semester name: Sem-6  
Number of courses: 2  
Disk usage: 56K .  
Directory path: /home/29061/runscript/Sem-6  
Created on: Sat 19 Sep 2026 01:32:04 SAST

```
(29061-@linux [2026-09-19 01:34:35])-[~/runscript/Sem-6] > cd Linux/
(29061-@linux [2026-09-19 01:34:48])-[~/runscript/Sem-6/Linux] > ls
Assignments Notes Revisions Solutions
(29061-@linux [2026-09-19 01:34:51])-[~/runscript/Sem-6/Linux] > cd ../OOP/
(29061-@linux [2026-09-19 01:34:59])-[~/runscript/Sem-6/OOP] > ls
Assignments Notes Revisions Solutions
(29061-@linux [2026-09-19 01:35:01])-[~/runscript/Sem-6/OOP] >
```



## SDS script source code

```
#!/bin/bash

#Semester Directory Setup (SDS) is a script for helping me setup my semester
files container gracefully and faster, it interacts with me, it is less time
consuming because, some repeatedly directories I have been creating manual, now
they can be created under few seconds by just accepting it. I have used different
linux commands, such as, echo, mkdir, cd, pwd, cat, du. Those helped to
successfully complete the SDS. I also used control structure such as while loop,
if, and for loop, while loop used to ensure that directories aren't already exist
in current directory, if used to make decisions like create directories or not,
for used to iterate through the number of courses user will enter to ask user
provide name of the courses. I also append redirection operator to generate the
report of the semester.

echo -e "\n\n=====";
echo "      WELCOME TO SEMESTER DIRECTORY SETUP (SDS)";
echo -e "=====\\n\\n";

echo -n "Enter your name: ";
read name;

echo -n "Enter your semester, e.g Fall2026: ";
read semester;

#This while loop is most important in this, script for handling user inputs
similar to existing directories. If user/me enters existing directory, the SDS
does not create that directory, unless user retyped different directory name

while [[ -d "$semester" ]]; do
    echo -n "The directory '$semester' semester already exist, Enter different
semester name: ";
    read semester;
done

echo -e "Success!\\n";
mkdir "$semester";
cd "$semester";

#The variable to hold the semester path
```



```
path="$(pwd)";

semesterDir=("Tables" "Payments");

i=1;
for dir in "${semesterDir[@]}"; do
    echo "$i. $dir";
    (( i++ ));
done

echo -ne "\nCan SDS create above directories in $semester? Y/N: ";
read response;

#Validating user choice to decide whether SDS can create directories for the user
if the user chose yes or y the SDS instantly creates the directories for the user

if [[ "$response" == "yes" || "$response" == "y" || "$response" == "Y" ]]; then

    for dir in "${semesterDir[@]}"; do
        mkdir "$dir";

    done

#Display success message to the screen after directories successfully created.
echo -e "Success!\n";
else
echo -e "\n";
fi

echo -n "How many courses for this semester?: ";
read num_courses;

#Validate the user input check if entered values is actual number or invalid
number number like less than 1

while [[ ! "$num_courses" =~ ^[0-9]+$ ]] || (( num_courses <1 )); do
    echo -n "ERROR: Number can't be less than 0 or be letter. Try again: ";
    read num_courses;
done

#Creating an array of most directories appear in each course
```



```
courseDir=("Notes" "Assignments" "Solutions" "Revisions");

echo -e "\n";
for (( i=1; i<=num_courses; i++)); do
    echo -n "Enter a name for course $i: ";
    read course_name;

    #Telling user that provided course directory exist so user can provide
different directory name
    while [[ -d "$course_name" ]]; do
        echo -n "You have already created '$course_name' course directory, Enter
different course name: ";
        read course_name;
    done
    mkdir "$course_name";
    cd "$course_name";
    echo -e "\n";
    j=1;
    for dir in "${courseDir[@]}"; do
        echo "$j. $dir";
        ((j++));
    done

    #Validating user choice to decide whether SDS can create directories for the
user if the user chose yes or y the SDS instantly creates the directories for the
user

    echo -ne "\nCan SDS create above directories in $course_name? Y/N: ";
    read response;
    if [[ "$response" == "yes" || "$response" == "y" || "$response" == "Y" ]];
then

        for dir in "${courseDir[@]}"; do
            mkdir "$dir";

        done

        #Display success message to the screen after directories successfully
created.
        echo -e "Success!\n";
        else
        echo -e "\n";
        fi

        cd ../;
done
```



```
#Using append operator to generate report of the directory in file called info

echo -e
"\n\n*****" >>
info;
echo "      Your $semester semester directory setup processed successfully" >>
info;
echo -e
"*****\n" >>
info;

echo "Student Name: $name">> info;
echo "Semester name: $semester" >> info;
echo "Number of courses: $num_courses" >> info;
echo "Disk usage: $(du -sh)" >> info;
echo "Directory path: $path" >> info;
echo "Created on: $(date)" >> info;
cat info;

#End of the scripts
```